Three Ways to Sample Students

Unit 1 · Topic 1.11 · TODAY'S PROBLEM

Suppose a school with four grades of 300 students each wants to estimate nightly hours of sleep. Three teams propose samples of 50.

Method 1: Choose a random start among the first 24 names on the alphabetical roster, then take every 24th name.

Method 2: Put all 1,200 names in a hat, mix thoroughly, and draw 50 without replacement.

Method 3: Separate the roster by grade, then randomly choose 12 students from each grade, then choose two grades at random and add one more randomly chosen student from each, for 50 total.

A student says, *They all use randomness, so they are the same.*

Three proposed samples

#	Selection rule	Total
1	Every 24th	50
2	Names from hat	50
3	Names within grades	50

FIRST TAKE (5 min, on your sheet)

Would these plans give the school the same mix of students every time? Give one reason from the plans.

DOK 1 **(a)** Name each method: systematic random, simple random, or stratified random.

DOK 2 **(b)** Which method is an SRS of 50 students? Explain using what must have an equal chance, not just the word random.

DOK 3 **(c)** Suppose sleep differs a lot by grade. Choose a method and explain what it guarantees about the mix of grades. Contrast it with the other two methods, then respond to the student's claim that all three are the same.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.